BHSM Frame Dragging and Nonlinear Front Formation

An action-based continuation of Norman Carberry’s galaxy-envelope hypothesis

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Main result. The recorded polynomial eta action contains a state-dependent characteristic cone and a nonzero nonlinear steepening coefficient in a specified scalar restriction. Suitable compressive simple waves can develop a gradient catastrophe while the local kinetic matrix stays uniformly positive. This supplies an action-based candidate for the formation stage of the proposed envelope.

Physical scope. This is a calculation in a prescribed-background matter sector. It is not a coupled galaxy solution, a completed admissible shock, a persistent BHSM boundary, or an explanation of an observed dark-matter discrepancy.

Obtained in this note	Result and limitation
Frame-dragging entry	The relative frequency is (nu - m omega)/alpha. A coordinate rotation alone changes no invariant.
Nonlinear propagation	The eta block has a modified cone. Its regular local branch is not a slow mode that stars overtake.
Steepening before kinetic failure	A nonzero directional derivative of characteristic speed gives a finite-time compressive simple-wave example.
Boundary support	The isolated source-free scalar restriction has no nontrivial stationary derivative jump on a connected positive-kinetic branch.
Galaxy gravity	No physical radius, lifetime, halo amplitude, metric slip, or galaxy fit has been calculated.

The distinction throughout is between an exact deduction from the specified sector and a physical identification requiring the rest of BHSM. No new empirical coefficient is fitted. The hypothetical frame is the prescribed parent/environment; interface and internal responses would have to follow from the action.

1 Source action and the frame-dragging differential

The August polynomial triality-spinor note [S1] and unified architecture [S2] supply the sector below. The September constrained child formulation [S3] retains the same polynomial in X, but uses a localization factor 1 - 4 sigma squared and a KKT response constraint. Its sigma dynamics cannot be replaced by the older independent scalar equation without a separate reduction.

$$\mathcal{L}_\eta = -f(\sigma)F(X), \qquad F(X) = \frac{\kappa_1 X}{2} + \frac{X^4}{8}$$

$$f(\sigma) = 1 + g\sigma^2 > 0, \qquad X = \langle D_A\eta, D^A\eta\rangle, \qquad |\eta|^2 = 1$$

Use signature (-,+,...), a connection preserving the positive internal inner product, and kappa1 greater than zero. The algebraic unit constraint is enforced by its multiplier. The normalization of this action is inherited as a structural postulate; it is not newly derived here.

Rotation enters the actual kinetic invariant

$$ds^2 = -\alpha^2 c^2 dt^2 + h_{\phi\phi}(d\phi - \omega dt)^2 + h_{AB}dx^A dx^B$$

$$D_\perp = \frac{D_t + \omega D_\phi}{\alpha c}, \qquad X = |D\eta|^2_{\text{spatial}} - |D_\perp \eta|^2$$

For a phase component with dependence exp(-i nu t + i m phi), with any gauge contribution treated in the actual covariant derivative, the corresponding local frequency is

$$\nu_{\text{loc}} = \frac{\nu - m\omega}{\alpha}$$

The background omega changes the temporal contribution to X. This is a concrete entry point for the proposed differential. An azimuthally trivial field with D_phi eta = 0 has no direct omega D_phi eta term. Coupled changes of its geometry or other fields can still matter.

$$\phi' = \phi - \Omega_0 t: \quad \omega' = \omega - \Omega_0, \quad \nu' = \nu - m\Omega_0$$

$$\nu' - m\omega' = \nu - m\omega$$

Thus a rigid coordinate relabeling cannot manufacture the differential. A physical comparison uses the specified environment, a consistent tetrad/reference convention, or relational curvature and field quantities. The metric is a local stationary axisymmetric comparison; no asymptotic infinity is imposed on the global hypersphere.

2 The eta principal symbol and kinetic domain

Set p(X) = 2F_X = kappa1 + X cubed, and V_A = D_A eta. Project variations onto the tangent space of the unit eta constraint. The eta equation and the second-derivative coefficient of its linearization are

$$\Pi_{\eta^\perp} D_A\left[fp(X)D^A\eta\right] = 0$$

$$P_{ab}(k) = f\left[p\,G^{AB}k_Ak_B\,\delta_{ab} + 6X^2 w_a(k)w_b(k)\right]$$

$$w_a(k) = V_a^A k_A$$

Indices a,b refer to an orthonormal internal tangent basis. The metric and connection are prescribed in this calculation. The rank-one update gives, for tangent dimension n,

$$\det P = f^n(pk^2)^{n-1}\left(pk^2 + 6X^2|w|^2\right)$$

$$C^{AB} = pG^{AB} + 6X^2\langle V^A, V^B\rangle$$

There is a metric-cone factor and one modified-cone factor. They are factors of this matter block, not a count of the physical modes of the fully coupled and gauge-reduced theory.

The inherited Legendre condition is recovered exactly

In a local orthonormal frame, let v = D_0 eta and X = sum_i |D_i eta| squared - |v| squared. The normal canonical momentum and velocity Hessian are

$$\pi_\eta = fpv, \qquad K_\eta = f\left[pI - 6X^2vv^T\right]$$

$$d = p - 6X^2|v|^2 > 0$$

With f positive, d positive makes the tangent kinetic matrix positive, implies p positive, and gives a Lorentzian modified form with a common admissible time slicing. This is a regular local propagation domain. Positivity of this block is not a certificate for the complete constrained theory.

The cone is not slowed in this sector

$$C^{AB}k_Ak_B = 0 \quad\Longrightarrow\quad G^{AB}k_Ak_B = -\tfrac{6X^2}{p}|w|^2 \le 0$$

The modified wave-normal speed is at least the metric light speed on this branch. Increasing inertia cannot be assumed to lower this particular propagation speed. A zero of d is a kinetic/evolution problem to diagnose; it is not automatically an encapsulation event.

3 A scalar restriction with calculable steepening

Prescribe a flat local metric, trivial connection, and constant positive f. Restrict eta to a great circle, eta = (cos q, sin q, 0, ...). This consistently reduces the unit-vector matter equation to one scalar; it does not solve the omitted metric, connection, or sigma equations. Work in units c = 1.

$$\tau = q_t, \qquad \chi = q_x, \qquad X = \chi^2 - \tau^2, \qquad p = \kappa_1 + X^3$$

The scalar Euler equation and derivative compatibility are two conservation laws:

$$\partial_t(p\tau) - \partial_x(p\chi) = 0, \qquad \partial_t\chi - \partial_x\tau = 0$$

$$a = p - 6X^2\tau^2, \quad b = 6X^2\tau\chi, \quad h = \kappa_1 + 7X^3$$

The characteristic speeds and a chosen eigenvector normalization are

$$\lambda_\pm = \frac{b \pm \sqrt{ph}}{a}, \qquad r_\pm = (-\lambda_\pm, 1)$$

Take a regular patch with p > 0, ph > 0 and a > 0. Differentiate each speed along its own eigenvector. The resulting Lax coefficient measures how that wave changes its own propagation speed:

$$\Gamma_\pm = \nabla_{(\tau,\chi)}\lambda_\pm \cdot r_\pm = \pm\frac{2\mathcal{E}(\chi + \lambda_\pm\tau)^3}{p\sqrt{ph}}$$

$$\mathcal{E} = pp_{XX} - 3p_X^2 = 3X(2\kappa_1 - 7X^3)$$

This coefficient is generically nonzero. Its isolated zeros at X = 0 and X cubed = 2 kappa1/7 do not make the whole theory linearly degenerate. A single null wave q = Q(x-t) can remain on X = 0 and avoid this steepening route. A nonzero differential is therefore not sufficient by itself.

This replaces the arbitrary coefficient in the earlier toy model

The relevant nonlinearity now follows from the recorded polynomial F(X). Its value still depends on the actual background and the normalization of the mode. Studies of nonlinear scalar and gravitational theories provide the characteristic method [P1-P3]; they do not establish this BHSM background or a galaxy interpretation.

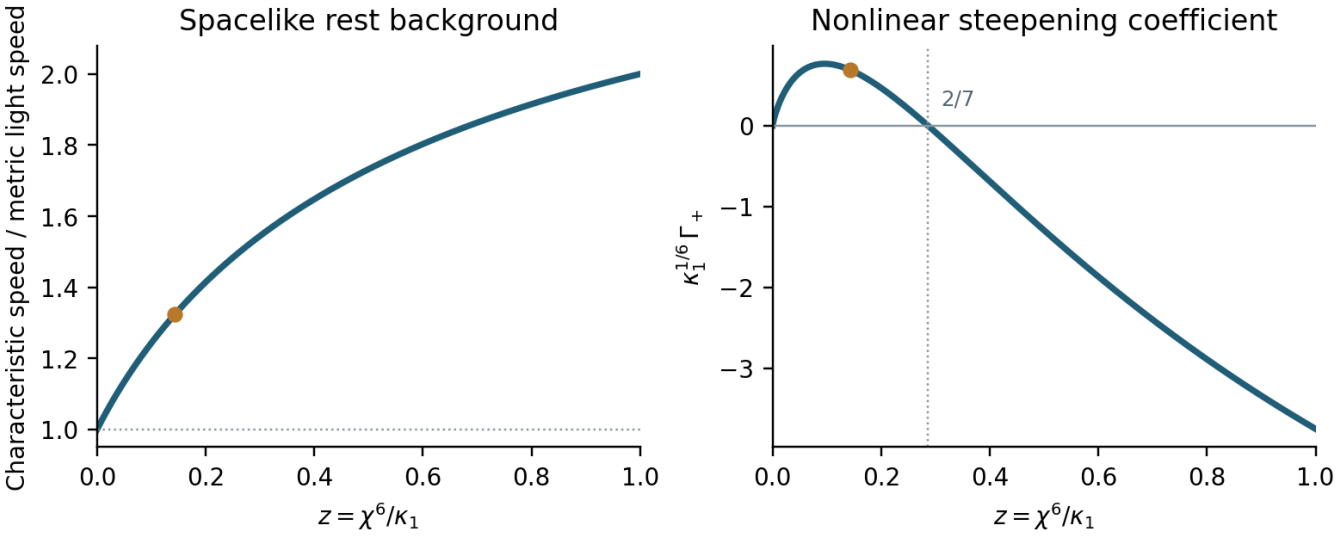
The eigenvector normalization fixes Gamma as written. Reparameterizing a simple-wave amplitude changes Gamma and the amplitude gradient together; their product, which determines the crossing time, is unchanged.

4 A regular example and finite-time formation

For a spacelike rest background tau = 0, choose chi positive with chi to the sixth power = kappa1/7. This is a dimensionless analytic example, not a fitted galaxy state. It lies strictly within the positive-kinetic domain.

$$p = a = \frac{8\kappa_1}{7} > 0, \qquad h = 2\kappa_1 > 0$$

$$\lambda_+ = \frac{\sqrt{7}}{2}, \qquad \kappa_1^{1/6}\Gamma_+ = \frac{3\,7^{2/3}}{16} \simeq 0.6861$$



Both panels use the frozen scalar restriction at tau = 0. Orange marks the example. Positive kinetic coefficients coexist with nonzero steepening. The zero at z = 2/7 is a leading nonlinearity zero, not kinetic failure. These are local field characteristics, not galaxy rotation speeds.

An exact simple-wave construction up to crossing

Choose an integral curve U(w) of r_+ through this regular state, with U = (tau,chi) and amplitude parameter w. Then U_w = r_+ and w obeys

$$w_t + \lambda_+(w)w_x = 0, \qquad x = \xi + \lambda_+(w_0(\xi))t$$

$$w_x = \frac{w_0'(\xi)}{1 + t\Gamma_+(w_0(\xi))w_0'(\xi)}$$

$$t_* = -\left[\min_{\xi}(\Gamma_+(w_0)w_0')\right]^{-1}$$

The time is finite when the indicated minimum is negative. Choose the initial state values in a small compact segment of the regular curve. Those values stay there up to crossing, while their gradients diverge. Thus a gradient catastrophe can occur before any loss of the kinetic margin. A physical formation history also needs t_* shorter than the available coherence and relaxation times.

5 How the differential can affect an envelope mode

This section uses the independent scalar sigma in the unified parent architecture [S2]. It does not substitute that scalar for the KKT-defined sigma in the later constrained child. Define Y = |d chi_field| squared, with the subscript distinguishing this carrier field from the scalar spatial derivative chi on the preceding pages.

$$\mathcal{B} = \frac{Z_\chi}{2} Y + F(X)$$

$$\mathcal{L} = -\frac{Z_\sigma}{2}(\nabla\sigma)^2 - U(\sigma) - (1 + g\sigma^2)\mathcal{B}$$

$$Z_\sigma \nabla_A \nabla^A \sigma - U'(\sigma) - 2g\mathcal{B}\sigma = 0$$

At a constant reference sigma = 0 with U prime(0) = 0, the quadratic coefficient is

$$\mathcal{C}_\sigma = U''(0) + gZ_\chi Y + g\kappa_1 X + \frac{g}{4} X^4$$

$$\partial_\omega \mathcal{C}_\sigma = gZ_\chi \partial_\omega Y + gp(X)\partial_\omega X$$

Frame dragging affects X and Y through the relative normal frequencies. This gives an actual sensitivity to calculate. Its sign is not universal. At sigma = 0 every displayed coupling is even in sigma: the background remains an exact solution, and dragging changes its stability rather than creating an inhomogeneous linear source. An instability needs a nonzero perturbation to grow.

Spatial extent and persistence enter the stability test

In a local static flat test with admissible compact perturbations, the sigma stiffness operator is H_sigma = -Z_sigma partial_x squared + C_sigma(x). If C_sigma is at most -mu squared on an interval of width L, a sine trial function yields

$$\lambda_{\min}(H_\sigma) \leq \frac{Z_\sigma \pi^2}{L^2} - \mu^2$$

Consequently mu squared L squared > Z_sigma pi squared is a sufficient instability condition for that test problem. A negative value at a single point is not sufficient. If an eigenvalue lambda is negative, the linear growth rate in c = 1 units is sqrt(-lambda/Z_sigma); appreciable growth requires a sufficient duration. This is a stability route distinct from the eta shock criterion.

At fixed total angular momentum the correct Legendre reduction and redistribution must be included. Since the displayed prefactor has first derivative zero at sigma = 0, matched fixed-frequency and fixed-momentum rotor curvatures agree there; switching ensemble cannot invent a quadratic sign reversal.

6 What a moving front must conserve

The scalar restriction supplies necessary matching conditions, but no prescribed physical rule for continuing after the gradient catastrophe. Let a planar front move at speed s, use brackets for right minus left, and assume q itself is continuous. Keep f constant and positive.

$$[\tau] + s[\chi] = 0$$

$$[p\chi] + s[p\tau] = 0 \quad \Longleftrightarrow \quad [p(\chi + s\tau)] = 0$$

The first equation is derivative compatibility; the second is continuity of canonical normal flux. If no surface stress or unresolved energy carrier exists, the scalar stress-energy tensor must also satisfy the normal conservation law:

$$T^{\mu\nu} = fp\,\partial^{\mu}q\,\partial^{\nu}q - fF(X)g^{\mu\nu}$$

$$n_{\mu}[T^{\mu\nu}] = 0, \qquad n_{\mu} = (-s, 1)$$

$$[F] = k[\chi], \qquad k = p(\chi + s\tau) \text{ common across the front}$$

This energy condition follows after applying the two earlier jump equations; it is an additional restriction, not an automatic consequence of them. A dissipative weak solution of the reduced equations requires the lost resolved energy to appear in other physical degrees of freedom. A dispersive completion can instead form an oscillatory transition. None is selected merely by naming the event a shock.

A stationary jump is excluded in this isolated sector

At s = 0, continuity gives equal tau and equal p chi. At that fixed tau,

$$\frac{\partial(p\chi)}{\partial\chi} = p + 6X^2\chi^2 > 0 \qquad (p > 0)$$

The flux is strictly increasing on a connected positive-p interval. Equal flux therefore gives equal chi, so both derivative jumps vanish. There is no nontrivial stationary source-free scalar derivative jump of this kind.

This is a precise limitation of the isolated restriction. It does not exclude a moving front, a smooth finite-width interface, or a boundary supported by the coupled eta, sigma, metric and interface stresses. The full theory must determine which of those realizes the proposed envelope. A bow-shaped front also does not automatically imply a closed enclosure.

7 Consequences for the galaxy hypothesis

The proposed causal route can now be assigned concrete quantities: a rotational source changes the background normal derivatives; these change X and the local characteristic matrix; a correctly signed compression can steepen; the complete coupled system must then select and support the interface. The front has to be formed by the galaxy/environment configuration, not supplied as arbitrary initial data when claiming a galaxy prediction.

Physical question	Object now identified
Where does dragging enter?	The covariant normal derivative and the relative frequency nu - m omega.
Can the recorded eta action steepen?	Yes, in the prescribed scalar restriction: Gamma is generically nonzero and a regular compressive example is given.
What selects formation time?	The initial/driven profile and min(Gamma times its gradient), together with coherence and relaxation times.
What supports the boundary?	The coupled field/interface solution and its flux, energy, domain and persistence conditions.
What produces the apparent halo?	The source exported by that same solution into the gravitational/topographic equations and the two metric potentials.

The earlier radial result remains downstream

The preceding galaxy note [S4] assumed a spherical exterior source QS_enc = A W/r squared, with W = GM_b/r_star and constant A. Under those assumptions its integrated contribution is

$$g_{\mathrm{enc}}(r) = AW\left(\frac{1}{r} - \frac{r_*}{r^2}\right)$$

$$v_c^2(r) = A\frac{GM_b}{r_*} + (1-A)\frac{GM_b}{r}$$

The present eta steepening coefficient is neither A nor the critical root xi_c. No identity connecting them has been derived. A confined shell or a shock location alone does not supply the assumed extended radial source. The complete configuration must determine its stress, spatial field, normalization and lensing response without counting the same source twice.

The ordinary weak-field gravitomagnetic contribution is already included in a mass-current integral and is small in conventional baryonic disk calculations [P4]. The proposed BHSM transition is an additional dynamical claim. No observed rotation curve was used to tune this note.

8 The concrete result and the remaining calculation

Established here, conditionally on the declared sector: the eta principal symbol, its Legendre matrix, the modified characteristic speeds, the nonlinear directional derivative, a regular compressive simple-wave construction, necessary moving-front conservation laws, and the absence of a stationary scalar derivative jump on the stated branch.

Also established in the older parent scalar sector: the direct coefficient through which a dragging-dependent kinetic invariant changes sigma stability, the absence of a linear sigma source at the symmetric state, and a finite-width variational instability test under explicit local assumptions.

The next mathematical problem is now specified

Construct a galaxy-scale admissible background with the baryonic current and stresses, prescribed external environment, action-selected internal/interface response, and the actual reduction of the retained BHSM fields. On that background, eliminate gauge and constraint directions and evaluate the coupled characteristic and nonlinear transport operators. The already derived frozen-block formulas are checks on that construction, not replacements for it.

Evolve a physically generated compressive profile to its transition. Resolve the front with the same action or a stated physical completion, enforce energy-momentum and interface balance, and determine whether the configuration persists. Its resulting stress and metric then determine the apparent mass discrepancy and lensing prediction.

Three thresholds must remain distinct

Threshold	Meaning
Characteristic crossing	Smooth evolution forms a divergent gradient in the chosen mode.
A stability eigenvalue crosses zero	A reference configuration can become unstable to an admissible perturbation.
A kinetic matrix loses rank	The selected evolution formulation loses its regular kinetic margin; this is not by itself a physical boundary.

These events can interact, but they are not interchangeable. The regular simple-wave example demonstrates that characteristic crossing does not require kinetic failure. The stationary-jump calculation demonstrates that front formation does not automatically supply boundary persistence.

9 Sources, assumptions and verification

BHSM source record

[S1] Norman Carberry research archive, BHSM v2.1 - Polynomial Triality-Spinor Stabilizer, bhsm_v2_polynomial_triality_spinor_stabilizer.md, 1 August 2026. Supplies the polynomial eta action, its unit constraint and tangential field equation. Its historical completion claims are not independently certified by this note.

[S2] Norman Carberry research archive, Unified Envelopment Architecture, bhsm_norman_unified_envelopment_architecture.md, 1 August 2026. Supplies the independent chi/sigma kinetic terms and the 1 + g sigma squared parent coupling.

[S3] Norman P. Carberry, BHSM Flagship Compression Derivation v0.3, 16 September 2026. Supplies canonical envelope terminology and the later constrained child with localization 1 - 4 sigma squared, KKT response and Hopf inertia. The cross-scale common-action identification is explicitly unfinished in that source.

[S4] BHSM Galaxy Envelope Formation and Persistence v0.1, 21 October 2026, preceding note in this conversation. The radial source export and persistence polynomial there are conditional constructions. This supplement advances the formation question; it does not convert those assumptions into predictions.

Primary literature for method and physical comparison

[P1] H. S. Reall, N. Tanahashi and B. Way, Shock Formation in Lovelock Theories (2014). Characteristic and nonlinear-transport methods; theory dependence of shock formation. arxiv.org/abs/1409.3874.

[P2] E. Babichev, Formation of caustics in k-essence and Horndeski theory (2016). Characteristic crossing and the need to address the continuation. arxiv.org/abs/1602.00735.

[P3] N. Tanahashi and S. Ohashi, Wave propagation and shock formation in the most general scalar-tensor theories (2017). Background-dependent cones and nonlinear discontinuity transport. arxiv.org/abs/1704.02757.

[P4] L. Ciotti, On the rotation curve of disk galaxies in General Relativity (2022). Ordinary weak-field mass-current corrections in baryonic disk models. arxiv.org/abs/2207.09736.

Verification and exact limits

The characteristic determinant, Legendre matrix, scalar directional derivative and matching equations were derived independently and compared. Six centered finite-difference checks of the characteristic directional derivative, covering spacelike, timelike and mixed regular states, agree with the analytic expression to better than 2 x 10^-8 in the chosen dimensionless examples. The marked analytic example is checked separately. These checks verify algebra; they are not empirical tests or a numerical solution of BHSM.

Assumptions: prescribed metric/connection for the eta block; positive internal pairing; positive prefactor; declared hyperbolic branch; a great-circle matter restriction with constant prefactor for the simple-wave calculation; distinct sigma sectors; no imported physical galaxy normalization; no assumed dissipative or dispersive completion. Physical causal consistency, full coupled constraints, finite-width continuation, envelope lifetime and observable gravity remain unevaluated.